Open problems: Equivalence of sup-norm, L^1, and L^2 seminorms on the Schwartz space

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Finite moments imply Schwartz.** "Finiteness of all such expectation values is equivalent to membership in $\mathcal{S}(\mathbb{R})$", and $\mathcal{S}(\mathbb{R})$ "is exactly the space of states for which all polynomial moments of position and momentum are well-defined". Only the bound (10) is formalized. The converse needs a precise statement: which orderings of X^α and P^β count, and what "well-defined" means for f outside their domains, since $(\|f\|^{L^2}_{\alpha,\beta})^2 = \langle X^\alpha \partial^\beta f | X^\alpha \partial^\beta f \rangle$ is not of the form $\langle f | X^\alpha P^\beta | f \rangle$. (Brief, Abstract, Remark after eq. (10) and Sec. 4; log, "Differences from the brief": not formalized.)

2. **General L^p seminorms.** "The same arguments extend to L^p seminorms for any $1 \leq p \leq \infty$", and all L^p-based Schwartz seminorm families generate the same topology. Not formalized. (Brief, Remark "General L^p"; log, "Differences from the brief": not formalized.)

From the formalization

1. None found.